Experiment 3

Thermodynamic Processes

Objective	Materials Used
Perform the three basic thermodynamic processes and complete an engine cycle	Heat Engine Apparatus Pressure gauge, rotational apparatus, temperature baths and thermometers

Background

Historically, the Ideal Gas law was first investigated in parts. Charle's Law (1780s) and Boyle's Law (17th century) examine the individual relationship between the state variables: Pressure, Volume, and Temperature. Charle's Law makes the following statement:

$$V = aT$$

This statement assumes that if the pressure of a system is held constant, the volume and temperature are directly proportional to each other. Boyle's Law makes the following statement:

$$PV = b$$

Under the condition that the temperature is held constant, the pressure and volume are inversely proportional to each other. By combining these equations together, the more general form of the ideal gas law can be found.

$$\frac{PV}{T} = d$$

Using the ideal gas law, our three basic thermodynamic processes can be examined. Isobaric refers to a system changing under constant pressure, isochoric refers to a system changing under constant volume, and isothermal refers to a system changing under constant temperature.

Apparatus

The apparatus is a general tool that can hold each of our state variables constant at a time. A release valve allows for the pressure to be held constant, the piston can be locked

Figure 3.1: Heat Engine Apparatus

to ensure the volume does not change, and the cylinder can be placed on a temperature bath to ensure the temperature does not change.
Connect the pressure sensor to the system as shown by your instructor. The volume is measured with the rotational apparatus with a mass hanging over the end to apply tension. The pressure and volume sensors connect to your computers to record measured values.

Procedure

1. Charle's Laws: $V = aT, P = const.$
   (a) Place the heat engine on its side to ensure the only force against the piston is from air pressure. Connect the cylinder to the piston through the release value and close the valve.
   (b) Place the metal cylinder in a water bath of around 80 °C. Measure the temperature and volume of the system.
   (c) Slowly add ice to the water bath and measure the temperature and volume every 5 °C until the temperature reaches 15 °C.
2. Boyle's Law: $PV = b, T = const.$
   (a) Stand the apparatus up and connect the pressure sensor through the release valve. Move the top of the piston to the upper most position and lock in the volume. Confirm that the pressure is atmospheric pressure and close the valve.
   (b) Unlock the piston and slowly apply pressure on top of the system (this can be done with you hand or by adding masses). Measure the pressure as the volume decreases in steps of 5 mm.

3. Heat Engine Processes: Isobaric, Isochroic, Isothermal
   (a) Prepare two temperature baths, one around 10 °C and the other around 60 °C. Connect the pressure sensor and temperature sensor using the split tube. Ask your instructor before continuing to ensure the setup is correct.
   (b) Review with your group how to perform the measurements for each thermodynamic process. Check with your instructor to ensure your plan is correct.
   (c) Connect the pressure and volume sensors to the Pasco Capstone interface. While you perform the three thermodynamic processes, create a PV diagram with the live data. Measure the work and heat based on your graph.

Analysis

1. Charle's Laws
   (a) Create a plot of Volume vs Temperature. Confirm this relationship is linear.
   (b) From this graph, measure absolute zero,
2. Boyle's Law
   (a) Create a plot of Pressure vs inverse Volume. Confirm that this is a linear relationship.
   (b) Confirm the value of your y-intercept makes physical sense.
3. Heat Engine Processes: Isobaric, Isochroic, Isothermal
   (a) With your PV diagram, note which part of the graph belongs to each thermodynamic process. Does this PV diagram follow the ideal for each process?

Conclusion

State if you agrees or disagrees with Charle's Law and Boyle's Law. Use your data and graphs to support this statement. Compare your measured value for absolute zero to the accepted value.